

Eren Dogan

<https://edogan.us>
erdogan@ucsc.edu

PhD Candidate in CSE at UCSC — VLSI/EDA & Graph ML — 4+ years of experience in hardware research & programming

EXPERIENCE

PhD Researcher

UCSC

Advisor: Prof. Matthew Guthaus

September 2022 – Present

- **GAT-Steiner: Rectilinear Steiner Minimal Tree Prediction Using GNNs**
 - Graph attention network model to predict the RSMT of any degree
 - 99.85% accuracy on the ISPD19 benchmark
 - Only 0.48% wire length increase on suboptimal predictions
 - Accepted to ICCAD 2024 (24% acceptance rate)
- **Router for OpenRAM**
 - Implemented a new supply and signal router for OpenRAM
 - Replaced the old, error-prone, grid-based router
 - Uses Hanan-grid graphs to align pins and wires perfectly
 - Faster, more precise, DRC-safe, and uses less wire

Teaching Assistant

UCSC & Ozyegin University

September 2019 – Present

- Assisted the following courses: Introduction to Data Structures and Algorithms (CSE 101), Introduction to Analysis of Algorithms (CSE 102), Computer Programming (CS 101), Digital Systems (EE 203), Computer Architecture (CS 240)

Research Intern

Ozyegin University

Advisor: Prof. H. Fatih Ugurdag

October 2020 – September 2021

- **OpenCache**
 - Open-source custom cache generator using OpenRAM's SRAM arrays
 - Outputs synthesizable Verilog file, configuration to run OpenRAM and obtain SRAM arrays
 - Simulates and verifies generated caches
- **Deep Compression for PyTorch Models**
 - Improved the "PyTorch to C generator" of Mr. Hasan Unlu of Tesla
 - Pruning and quantization applied on PyTorch models
 - Decreases memory usage and improves speed with compressed sparse column (CSC) format

EDUCATION

University of California, Santa Cruz

PhD in CSE; GPA: 3.86/4.00

Santa Cruz, CA

September 2022 – Present

Ozyegin University

BS in CS; GPA: 4.00/4.00

Istanbul, Turkey

September 2017 – June 2022

Honors: Valedictorian, Full-tuition scholarship

PROGRAMMING SKILLS

Languages: Python, C/C++, (System)Verilog, Chisel, Java, JavaScript, SQL, Bash

Tools & Frameworks: Linux, Git, PyTorch, PyTorch Geometric, Tensorflow, Spektral, Amaranth, Makefile

SERVICE

Reviewer: DAC 2025, MLCAD 2024, VLSI-SoC 2024

Volunteer: MLCAD 2025

Member: IEEE